

Information Computer Systems and Python Programming

Midterm

Part II. Python

Instructions:

Create a .ipynb or .py file according to a template

<Name>_<Surname>.ipynb/.py. Each task (including the bonus ones) must be completed in a separate function called **func_<task number>**. The .ipynb/.py file with your solutions should be uploaded to the folder D://Exam_solution on your computers. You are strictly prohibited from using any libraries, using one would result in automatic zero for the task.

During the exam you will have an opportunity to earn bonus points via solving the bonus problems, note that **your total score for the exam won't exceed 100 points even if you score more than that through solving bonus problems.** The python part overall can give you 60 points and additional 20 if you solve the bonus tasks, 1 bonus point per mandatory task can be earned if specifying the types of inputs and outputs of the functions in the mandatory tasks.

Task 1 (3 pts)

Given a list **arr** increment all of the values by 5 and multiply them by 10 afterwards

Examples:

Inputs: [1, 2, 3, 4]

Outputs: [60, 70, 80, 90]

Inputs: [5, 8, 9, 11, 16, 80]

Outputs: [100, 130, 140, 160, 210, 850]

Inputs: [0.8 , 0.06, 0.76, 0.95, 0.1]

Outputs: [58.0, 50.6, 57.6, 59.5, 51.0]

You can use the **round** function to round the floating values to some precision

Task 2 (5 pts)

Given a list **arr** and a number **x** change the values in the **arr** that are equal to **x** to zero, otherwise increment them by 10 and then divide them by 5.

Examples:

Inputs: [1, 2, 3, 4], 3

Outputs: [2.2, 2.4, 2.8]

Inputs: [10, 3, 11, 42, 54], 42

Outputs: [4.0, 2.6, 4.2, 12.8]

Task 3 (7 pts)

Given two arrays **arr_1** and **arr_2** return the sum of all values that are both equal numerically and stand on the same index in the array.

Examples:

Inputs: [1, 4, 0, 2, 9, 9, 0, 3, 6, 7], [8, 9, 4, 7, 0, 2, 1, 7, 6, 3]
Outputs: 12

Inputs: [1, 0, 1, 2], [0, 3, 0, 2]
Outputs: 4

Inputs: [1, 1, 3, 1, 3], [1, 3, 3, 0, 0]
Outputs: 8

Task 4 (10 pts)

Given two arrays of numbers **arr_1** and **arr_2** return the array that is the result of elementwise division between the two arrays, if the division can't be performed on at least one pair of values, signify the issue to the user by printing the error message, put a None value in the place of the result of the division.

Examples:

Inputs: [1, 1, 3, 1, 3], [1, 3, 3, 0, 0]
Outputs: [1.0, 0.3333333333333333, 1.0, None, None]

Inputs: [6, 8, 3, 4, 9], [3, 0, 2, 2, 1]
Outputs: [2.0, None, 1.5, 2.0, 9.0]

Inputs: [5, 8, 1, 7, 9], [3, 0, 3, 1, 3]

Outputs: [1.666666666666667, None, 0.333333333333333, 7.0, 3.0]

Task 5 (15 pts)

No arguments are initially given. You should ask the user to input some number **n**, then ask them **n** times to input some word. The request for the input of the word must contain the index of the current request. Return the list of words input by the user.

Example:

Inputs:

2

Please input word number 0: Hello

Please input word number 1: World!

Outputs: ["Hello", "World!"]

Task 6 (20 pts)

No arguments are initially given. You are collecting data on people and this is your to-be survey. You should ask the user to input their name, sex and age. The function is expected to keep asking for the input until the user inputs "STOP", then the program must write this data into some file, if the file exists, it can be overwritten (or not, it's up to you), if it doesn't exist- it should be created. No return is expected.

Example:

Inputs:

Enter your name: Ivan

Enter your sex: Male

Enter your age: 20

Enter your name: Oleg

Enter your sex: Male

Enter your age: 54

Enter your name: STOP

Outputs: None

Task 7 (bonus, 4 pts)

Given an array **arr** remove all of the duplicate values in the array, return the number of unique values in **arr**.

Examples:

Input: [1,1,2]

Output: 2

Input: [3, 4, 2, 4, 0, 4, 0, 1, 2, 2]

Output: 5

Task 8 (bonus, 5 pts)

Given an array **prices** where **prices[i]** is the price of a given stock on the *i*th day. You want to maximize your profit by choosing a single day to buy one stock and choosing a different day in the future to sell that stock. Return the maximum profit you can achieve from this transaction.

Examples:

Input: prices = [7,1,5,3,6,4]

Output: 5

Input: prices = [7,6,4,3,1]

Output: 0

Task 9 (bonus, 5 pts)

Given an array **nums** containing n distinct numbers in the range[0, n], return the only number in the range that is missing from the array.

Examples:

Input: nums = [3,0,1]

Output: 2

Input: nums = [0,1]

Output: 2